Counterfactual Analysis Report

What-if scenarios showing how the anomaly could have been prevented

Original anomaly: CPU_SPIKE

Model confidence: 78.5%

Feasible prevention found: YES

Computation time: 120.5ms

Recommendation: Reduce overall workload or scale horizontally (add replicas)

Scenarios compared: 4

This report visualizes counterfactual scenario comparison, required intervention magnitude, and the best recommended prevention strategy.

Counterfactual Summary

Counterfactual Summary

Original anomaly: CPU_SPIKE (78.5% confidence)

Best scenario: cpu_utilization_mean

Current -> Target: 0.85 -> 0.40

Required change: -52.6%

Predicted outcome: NORMAL (52.9%)

Scenario score: 75.9/100

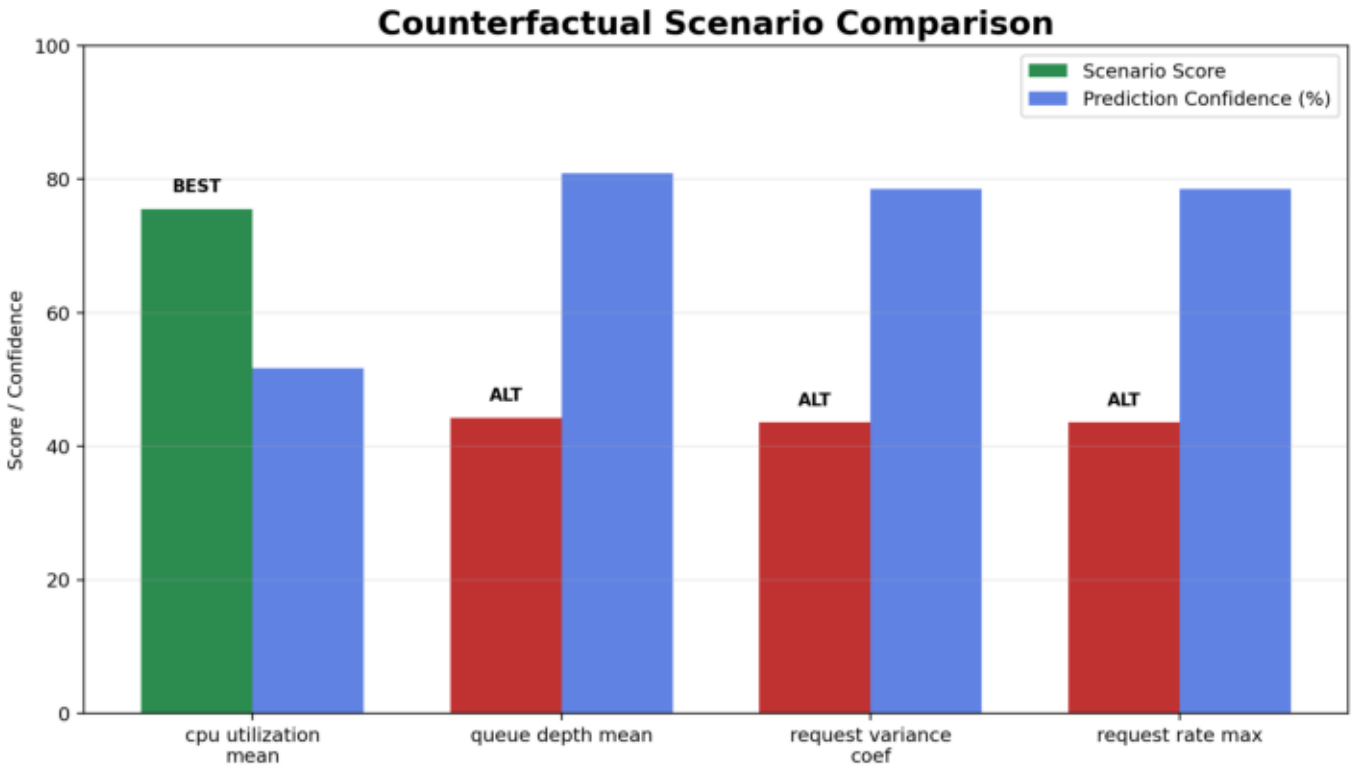
Risk level: MEDIUM

Compared alternatives: queue_depth_mean, request_variance_coef, request_rate_max

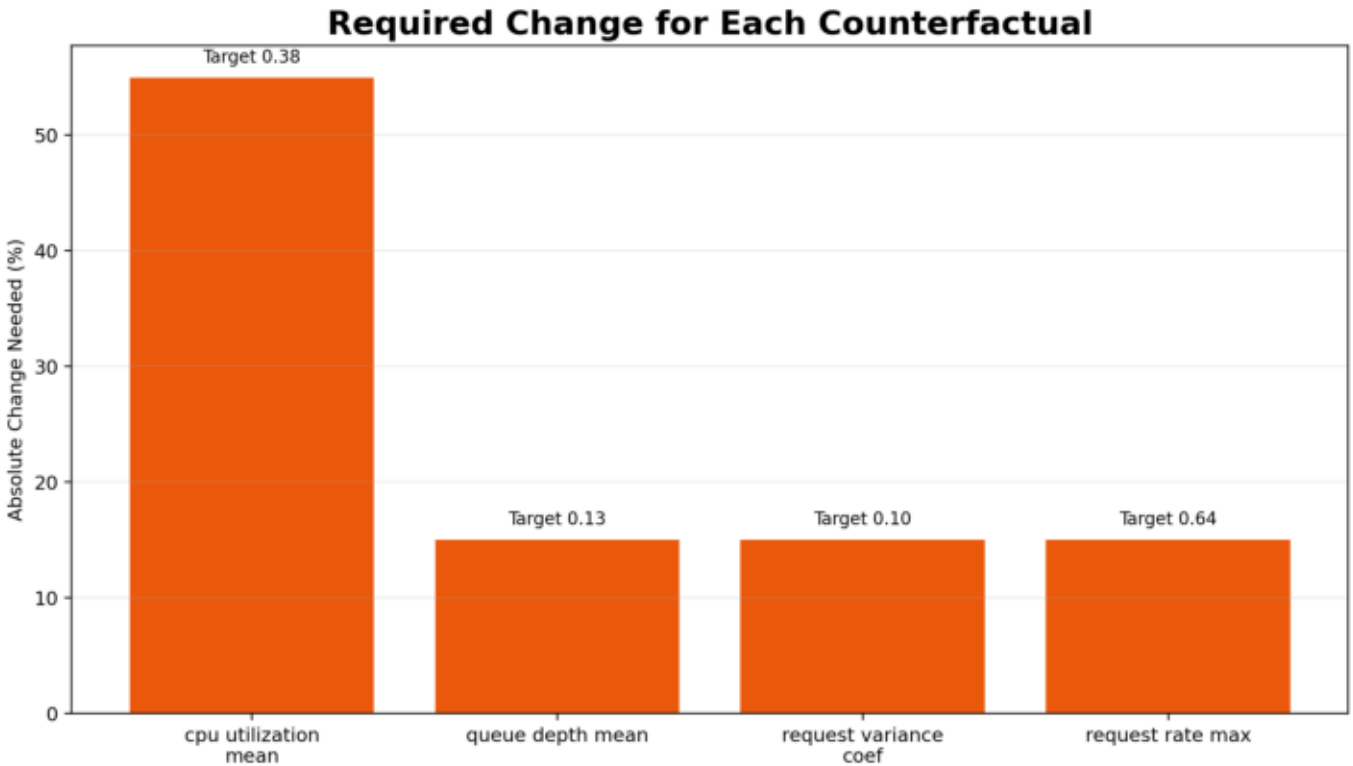
Recommendation: Reduce overall workload or scale horizontally (add replicas)

Computation time: 120.5ms

Scenario Comparison



Required Change Analysis



Detailed Counterfactual Findings

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Original anomaly: CPU_SPIKE (78.5%)

1. Scenario: cpu_utilization_mean
   Change: 0.85 -> 0.38 (-55.0%)
   Predicted outcome: NORMAL (51.7%)
   Prevents anomaly: YES
   Score / Risk: 75.5 / MEDIUM
   Best scenario: YES

2. Scenario: queue_depth_mean
   Change: 0.15 -> 0.13 (-15.0%)
   Predicted outcome: CPU_SPIKE (80.9%)
   Prevents anomaly: NO
   Score / Risk: 44.3 / LOW
   Best scenario: NO

3. Scenario: request_variance_coef
   Change: 0.12 -> 0.10 (-15.0%)
   Predicted outcome: CPU_SPIKE (78.5%)
   Prevents anomaly: NO
   Score / Risk: 43.6 / LOW
   Best scenario: NO

4. Scenario: request_rate_max
   Change: 0.75 -> 0.64 (-15.0%)
   Predicted outcome: CPU_SPIKE (78.5%)
   Prevents anomaly: NO
   Score / Risk: 43.6 / LOW
   Best scenario: NO

Best recommendation: Reduce overall workload or scale horizontally (add replicas)
Total computation time: 120.5ms
```